

OLIVER CARNEW

GRADUATE ELECTRICAL ENGINEER

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PROFESSIONAL SUMMARY

Graduate electrical engineer with applied experience in engineering consultancy, specializing in Python data pipeline automation and the quality assurance of machine-learning outputs. Demonstrated hardware design capabilities spanning embedded C/C++ firmware, bench instrumentation, and end-to-end PCB layout using KiCad and Altium. Proven portfolio of complex instrumentation builds—including an ECG front-end and an industrial data logger—coupled with Honours research quantifying neural model reliability under realistic variations.

EDUCATION

Bachelor of Engineering (Honours) — Electrical & Biomedical Engineering

Graduated 2025

The University of Queensland

- Honours thesis: evaluated the robustness of machine-learning search and ranking systems — built automated pipelines to process large text datasets and benchmarked traditional and neural models, quantifying a ~45% performance drop under realistic input variation.
- Completed team-based design-and-build project coursework throughout the degree, delivering working prototypes to specification and deadline.
- Focus areas: control systems, embedded systems, signal processing, instrumentation, machine learning.

TECHNICAL SKILLS

Programming: Python (NumPy, pandas, Jupyter), C/C++ (embedded), MATLAB/Simulink, JavaScript; familiar with SQL

Embedded & Hardware: Microcontroller firmware (C/C++), FreeRTOS, serial protocols (UART, SPI, I2C), schematic capture and PCB layout (KiCad, Altium Designer), hardware bring-up, bench test/debug (oscilloscope, logic analyser), instrumentation and data acquisition — portfolio projects include an ECG front-end, industrial DAQ data logger, battery and BMS, USB-PD power supply and an embedded rhythm game

Data & Machine Learning: Large-dataset processing and pipeline automation, neural model training, evaluation and benchmarking (TensorFlow), statistical analysis, data visualisation

Engineering Practice: Technical documentation (memos, user guides), quality assurance, version control (Git), Scrum/Agile, WHS and electrical safety awareness, stakeholder and client communication

ENGINEERING EXPERIENCE

Undergraduate Engineer (Intern) | Alluvium Consulting, Brisbane

Nov 2024 – Feb 2025

Three-month internship with a national environmental engineering consultancy; responsibilities included building and documenting an internal data-analysis workflow used to generate recurring client reports.

- Rebuilt and extended an open-source satellite-image analysis toolkit into a documented in-house workflow, automating data retrieval, machine-learning classification and post-processing to produce recurring client deliverables.
- Upgraded a core scientific model to its latest generation (FES2014 to FES2022), re-engineering data retrieval around the provider's new distribution service and validating outputs against legacy results.
- Quality-assured machine-learning classification outputs, tuning inputs and reviewing results so deliverables met client standards.
- Authored technical memos and programs enabling engineers and non-programmers across the team to run the full system independently, reducing key-person risk.
- Presented progress at weekly team meetings and worked daily with civil, water and software engineers to align tooling with project needs.

Virtual Work Experience | Engineers Australia

2023

Worked within simulated multidisciplinary engineering teams to deliver project outcomes, assessing RoHS compliance requirements and applicable governing standards.

Survey Assistant / AJS Surveys, Brisbane**2021 – 2023**

Operated and maintained precision survey equipment alongside licensed cadastral surveyors across residential and commercial sites.

- Prioritised measurement accuracy and data integrity, following survey procedures and verifying setups across varied site conditions.
- Held responsibility for the safety of contractors and bystanders around live survey operations.

Owner / Operator / OC Computer Solutions**2017 – 2021**

Founded and operated a computer solutions business serving small businesses and private clients.

- Designed, built and tested custom PC and network solutions to client requirements; managed repairs, callouts and diagnostics.
- Ran full business operations, including inventory, invoicing, client relations and A/B-tested marketing campaigns.

ADDITIONAL EXPERIENCE

Owner / Disability Support Worker / OC Support Services**2023 – Present**

Founded an independent NDIS support practice and grew it to full-time operation. Provide support for clients living with disability, ethically implementing Positive Behaviour Support Plans in coordination with psychologists and allied-health teams; manage NDIS invoicing, compliance, scheduling and support-team coordination. Certified in restrictive practices (QLD Health), medication assistance and manual handling.

Owner / Private Tutor / OC Tutoring**2014 – 2023**

Delivered personalised tuition in Mathematics (A/B/C), Physics, Engineering and English for Years 7–12, plus piano for all ages; managed client outreach, scheduling and invoicing.

Area Production Leader / Crew Supervisor / McDonald's Australia**2014 – 2023**

Led production teams of 2–6 crew, training new and existing staff and coordinating shift operations to meet quality, safety and food-hygiene standards.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Technical: Harvard CS50 Computer Science; SQL Developer Certification; IBM Text Analytics 101; Scrum Fundamentals Certified; Google Digital Marketing Fundamentals; Inbound Marketing Certification

Industry & Safety: White Card (WHS Construction); First Aid & CPR; Manual Handling Accredited Training; Restrictive Practice Certification (QLD Health); Blue Card (WWCC); Yellow Card (NDIS); Certificate II in Active Volunteering

Music: AMusA — Associate Diploma in Music, Piano (AMEB, 2016)

MEMBERSHIPS & INTERESTS

Memberships: Engineers Australia (Member); Engineering Undergraduate Society (UQ); Electrically Based Engineering Student Society

Interests: Outdoor team climbing, surfing, spearfishing, 4WD, piano

References available upon request